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Chronological independently verifiable electronic chain of custody ledger using blockchain technology

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ABSTRACT

As evidence in digital form becomes more prevalent in all kinds of investigations, there is a pressing need for a trustworthy and transparent way to maintain electronic chain of custody (e-CoC) and integrity information that is independently verifiable. Generally, a hash value of digital evidence is calculated and documented with the acquired data to prove that it has not been altered. However, the hash value alone does not prove that digital evidence is the same as when it was obtained, only that the contents has not been modified since the time when the hash was calculated. This work responds to the need for a chronological independently verifiable e-CoC ledger using blockchain technology. Employing this approach, each e-CoC record is stored in a block, each block being connected to the previous one with the hash value of the block. This e-CoC ledger, the blockchain, can be hosted by a trusted entity and accessed by any party to verify e-CoC details. For privacy reasons, sensitive information is not stored inside the e-CoC record in the blockchain. Moreover, to prove that the e-CoC ledger itself has not been modified, information is periodically sent to a public blockchain, where the integrity is guaranteed by its decentralization and the structure of such a secure ledger. Not all of the blocks are sent into a public blockchain which allows different levels of verification. Proof-of-work examples are provided using AFF4 and dc3dd to demonstrate how this e-CoC ledger can be used by different parties in a legal context, including digital forensic practitioners, attorneys and judges.

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1. Introduction

Decisions impacting criminal justice, security, health and governance are becoming increasingly reliant on information of probative value stored in binary form, namely digital evidence. For example, any type of criminal investigation can have a digital nexus: security breaches of organizations typically involve digital traces, eradicating serious health problems requires sharing digital data, and eGovernance depends on digital documents. In addition to data created using computers such as documents, multimedia, communications, and activity logs, traditional types of evidence such as DNA or drug samples processed using computers can be stored in digital form.

When reaching high stakes decisions on the basis of digital evidence, it is crucial to maintain provenance information in a manner that can be used to establish their authenticity and trustworthiness [Casey et al., 2015]. In a legal context, authentication involves demonstrating that digital evidence originated from its purported source, its contents has not changed since it was obtained, and associated details such as the apparent date of the record are accurate [Ćosić and Bača, 2010; Evans and Stagner, 2003; Richter et al., 2010]. Maintaining provenance information of digital evidence is particularly challenging given the ease with which it can be copied, transformed, and damaged. Multiple parties can work on digital evidence, increasing the risk of alteration, whether intentional or inadvertent [Ćosić and Ćosić, 2012]. Furthermore, each involved party has a need to independently verify the authenticity and integrity of digital evidence.

For forensic purposes, provenance information is maintained using Chain of Custody (CoC) documentation to keep track of handling of digital evidence [Schum et al., 2009]. The CoC includes details about who handled digital evidence when, where, how, and typically provides a way to verify the integrity of the data. The most common way to verify the integrity of digital evidence is using a

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cryptographic hashing function [Rasjid et al., 2017], enabling the fact-finder to verify that the evidence has not been altered by any handling or processing after it was obtained. Unless a secure tamper-evident mechanism is in place, there is nothing preventing a dishonest individual from subsequently altering the digital evidence and associated hash value. The fact-finder can only verify that the evidence has not been altered since the moment when the hash value was calculated. This particular temporal information is not stored within the hash itself, and timestamps associated with digital evidence can be backdated. To satisfy the growing demand for dependable, scalable, automatable authentication of digital evidence, there is a need for independently verifiable, chronological CoC ledger that includes details necessary to verify original data integrity.

> "With the advent of the digital age, the tangible CoC document needs to [undergo] a radical transformation from paper to electronic data (e-CoC), readable and consumed by machines, and applications." [Gayed et al., 2012]

This work presents a solution based on [Burri and Jaquet-Chiffelle, 2018] implemented using a trusted entity, which manages a secure ledger, as described by Haber and Stornetta, (1991) at the beginning of the digital era. Blockchain technology provides the ideal solution of this digital transformation of provenance information related to digital evidence. A blockchain is composed of multiple blocks and uses the hash of each block to link it to the previous one. Thus, no alteration can be executed without being evident: once data are stored in the blockchain, they cannot be modified or removed [Schär and Berentsen, 2017]. In the block there is the information on when the block was submitted into the blockchain. Before any block is accepted into the blockchain, it has to pass a proof-of-work, that any node in the network can do [Nakamoto, 2009]. Public blockchains are decentralized, allowing anyone to access the blockchain and observe what has been stored. A private blockchain only allows access to trusted parties, making it more suitable for use in criminal justice contexts and other situations involving sensitive information. Both public and private blockchains support scalability and automation – they do not have limits on the number of blocks that can exist and can be operated on programmatically.

An added level of assurance can be attained by regularly recording the state of this private e-CoC ledger into a public blockchain in order to protect the trusted entity itself [Bossuat-Pahud and Jaquet-Chiffelle, 2018]. The e-CoC ledger presented in this work reinforces rather than replaces existing chain of custody processes.

This paper begins with an overview of related work on improving the CoC followed by background on blockchains and how they work. The solution developed in this work is detailed and discussed, and integration with AFF4 and dc3dd are provided as a proof-of-work. Finally, attacks that could be performed against this e-CoC ledger are presented along with explanations of how they can be prevented.

2. Related work

Earlier efforts to enhance methods for assuring integrity of

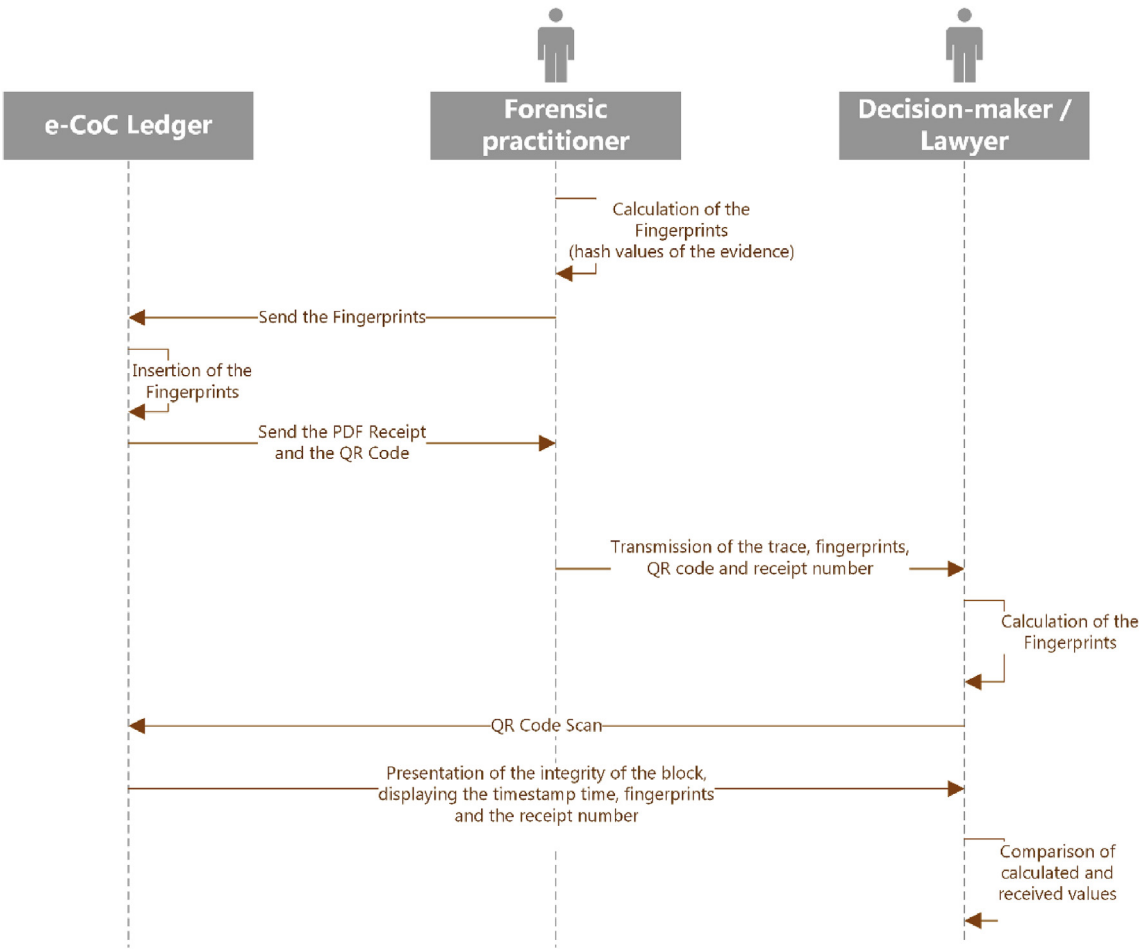


Fig. 1. Process of the use of the e-CoC ledger.

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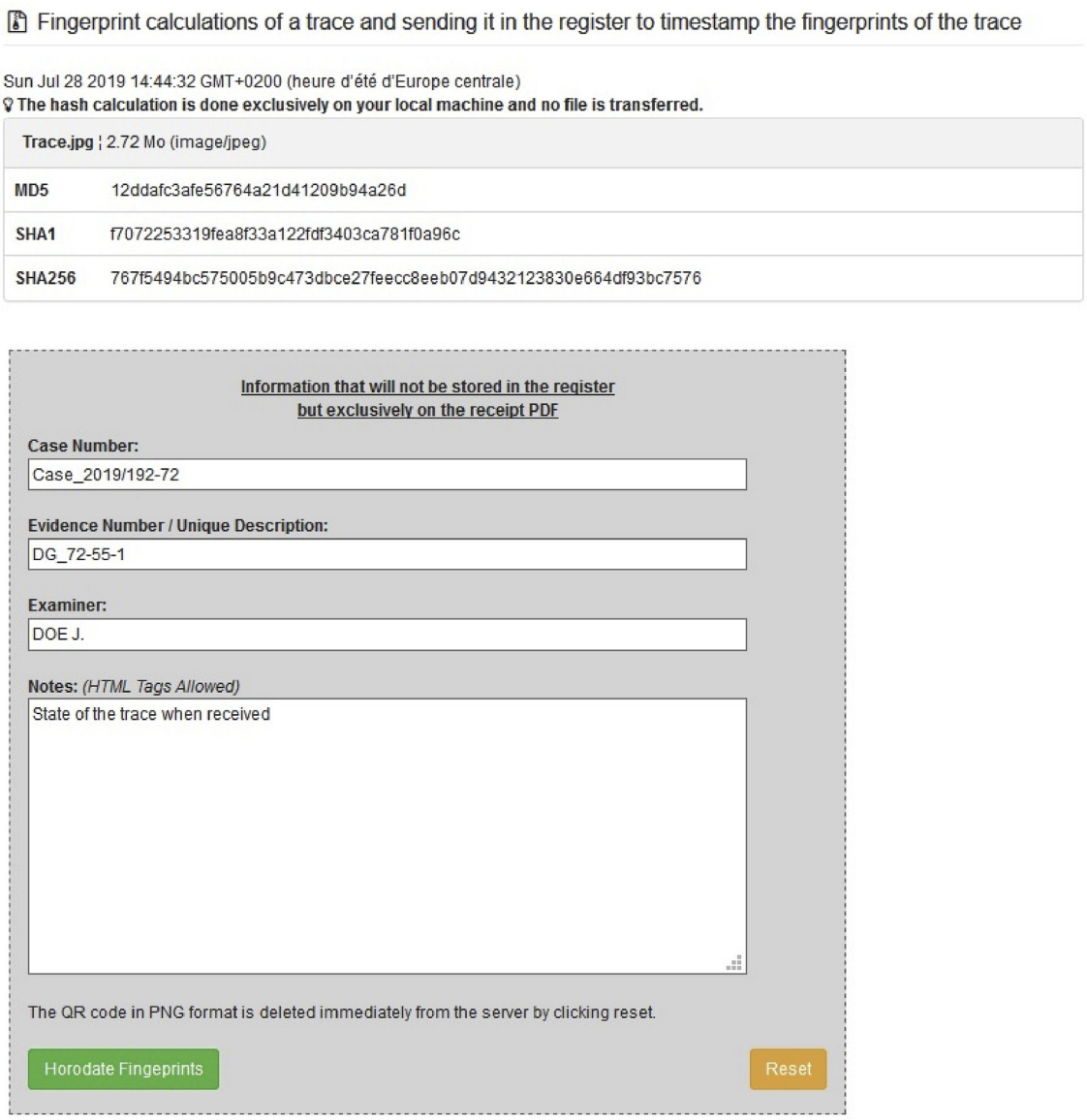


Fig. 2. Interface of the e-CoC ledger.

digital evidence used public key cryptography [Garfinkel, 2009]. The integration of digital signatures into the Advanced Forensic Format (AFF) strengths the link between the entity (device, person, or group) that acquired data from an evidential source. In addition, this solution adds authentication of both the acquired data and associated metadata. However, a private key used for signing can be compromised and the associated temporal information is only as reliable as the time source used.

A comprehensive approach to establishing and maintaining provenance information of digital evidence proposed using biometric information, cryptographic algorithms, GPS data and trusted time [Ćosić and Bača, 2010]. Using a time source that is reliable is certainly beneficial, and ideally should be independently verifiable. A CoC system that relies on biometric information, however, requires a database of everyone who will work with the data, which is infeasible in practice and raises data protection challenges.

Prior work on blockchain-based chain of custody, called Forensic-Chain, proposed an approach to auditable CoC throughout a digital investigation [Lone and Mir, 2019]. Although Forensic-Chain demonstrated useful concepts, its complexity makes it difficult to implement.

Any proposed approach to using blockchain technology to strengthen chain of custody should provide a proof-of-work that can be easily used by any person and tool throughout the digital investigation process.

3. Blockchain

The CoC ledger detailed in this work is based on blockchain technology, where each block is connected to the previous one, as Lone and Mir, (2019) also proposed. The use of blockchain in this context is beneficial for the CoC, because each block is connected with the parent one through the hash of the block [Zheng et al., 2017]. Thus, a deletion or an alteration of a block would be immediately discovered as the reference in the following block would be incorrect. However, unlike public blockchains, the implemented e-CoC ledger is not decentralized. Public blockchains are thus also used in this work to guarantee that the trusted entity, in charge of

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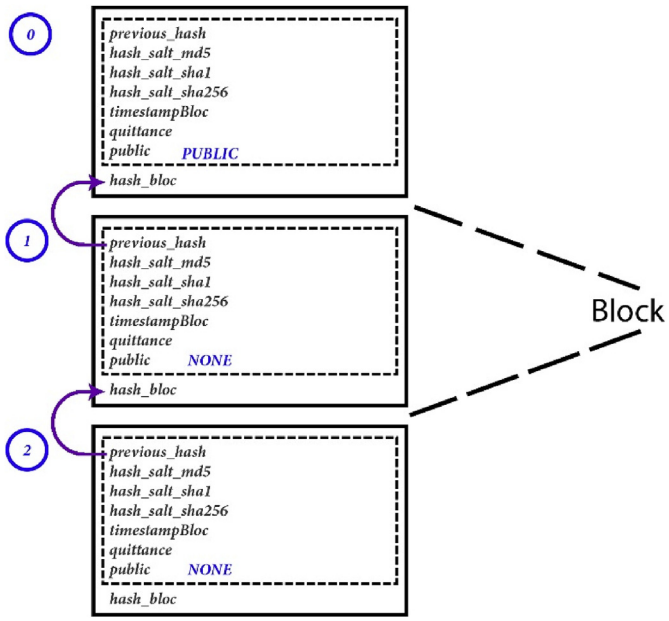


Fig. 3. Structure of the e-CoC ledger.

managing the e-CoC ledger, does not intentionally or accidentally modify blocks older than some chosen (acceptable) length of time, typically half a day or one day. Indeed, data registered in public blockchains cannot be altered, because they are, as said before, decentralized and everybody can verify the authenticity of the data [Lu, 2019]. Any alteration in a public blockchain would be immediately discovered. Thus, the public blockchains have a significant role to play in legal contexts requiring a more robust chain of custody.

4. Solution requirements

A fundamental requirement of any e-CoC solution is that it must be tamper-evident. In addition, an e-CoC solution must maintain the integrity of itself and of the data it contains. The e-CoC system must also be accessible by everyone who will require chain of custody information throughout a digital investigation. An e-CoC solution must also be scalable, either with the number of entries or with the added information it stores. To be more widely used, the solution should also be automatable, to facilitate the work of digital forensic examiners.

The solution must be user friendly. Specifically, its use should not be too difficult and work-consuming for the investigator. Moreover, as it is intended for lawyers and decision-makers as well as forensic practitioners, it cannot be highly technical and should stay as simple as possible.

As evidence can contain sensitive data and for privacy reason, the evidence itself should not be exposed in the e-CoC ledger. Thus, the evidence is not directly sent, but only the hashes of the evidence are used and salted hashes are stored.

5. Solution implementation

This work builds on prior work using a trusted entity to reinforce e-CoC by using public blockchain to protect particular blocks [Burri and Jaquet-Chiffelle, 2018]. The current implementation is compatible with an already-developed platform, which uses Django and JavaScript.

The process of using the interface to timestamp evidence is shown on Fig. 1. Although we speak only about timestamping digital evidence in this article, it would also be possible to timestamp pictures or even log files of evidence acquisition containing the hash of the evidence. Such a system is not only applicable to digital evidence but could also be used in other forensic disciplines, for all the digital documents or results, such as digital images and analytical data. It can be useful to any actor who needs to verify the e-CoC in a particular case, such as investigators, decision-makers, and lawyers, so they can verify when hash values of the evidence they receive were computed.

First, as soon as forensic practitioners obtain the evidence, they calculate its hash values directly on the local machine (in the browser in our implementation as shown in Fig. 2). In this way, no sensitive information is sent to the e-CoC ledger or to the trusted entity that manages the ledger. Then, the forensic practitioner can insert some metadata about the evidence and/or case. Those metadata will not be stored in the e-CoC ledger and are only used for the PDF receipt. Once a forensic practitioner sends the hash values to the ledger, only salted hashes will be stored in the ledger, as well as the time of the operation, meaning the timestamp of the hash. The e-CoC ledger will then send back the PDF receipt and a QR code to the forensic practitioner. The receipt contains the QR code, the non-salted hashes, the timestamp, the receipt number and the metadata about the case.

In a second step, the forensic practitioner gives the evidence to the court with the QR code and the hashes of the trace. The decision-maker will be able to recalculate the hashes of the trace and scan the QR code. The interface will show him/her the hashes


```
{
    "__previous_hash": "03de748821c91c701a3dd7b597e07064286726f928f4c740f7087b2d644f4fa4",
    "__hash_salt_md5": "d2f366ebbea9bf32b40464f4795289da5298e6d5666d9307536ad0a37f9b1a3b",
    "__hash_salt_sha1": "765f148d1cd412b79438a515d5693b590c45eaca1f3ecab826218f052e7e0eb4",
    "__hash_salt_sha256": "adc76392ff6acb43ba461fd84b6b2feb84536edc2760c679bb69d5197756c8b4",
    "__timestampBloc": "2018-09-02 18:50:42 (CEST +0200)",
    "__quittance": "3D29-4490-8DEC",
    "__public": "None",
    "__hash_bloc": "90ccc94ffab9470c37f49a40c15c2a36249a6cb2a6f2f666106461646d739657"
},
{
    "__previous_hash": "90ccc94ffab9470c37f49a40c15c2a36249a6cb2a6f2f666106461646d739657",
    "__hash_salt_md5": "641a74faff1b01c9c7b51ad602e8f1a631797d01f5e9c3e2673db4d373c9900f",
    "__hash_salt_sha1": "749a9532f3143da913e438217650b55fdccc432913aa786254ecfa46219795c3",
    "__hash_salt_sha256": "8d960a7758d3d789db52162fca651554138f4018f4c1301bd06f570edcc4a098",
    "__timestampBloc": "2018-09-02 18:51:28 (CEST +0200)",
    "__quittance": "3CB7-45BC-8F2A",
    "__public": "{\"blockchain2\": \"aeeb23b50bae4a77aec464c516cd2d6b\", \"Date_envoi\": \"2018-09-02 18:51:31\"}",
    "__hash_bloc": "6736b59a5b7aea305da00bffb80b72efcff16c0d5158a7ba09af860e658224db"
},
```


Fig. 4. Content of two blocks.



https://registre.[redacted]/search?md5=12ddafc3afe56764a21d41209b94a26d&p_hash=bce0551090804e1d3c47803a49af14bb7e5358bc1f079ec0973eb84b67136c62&salt=7d516d85a97c466e83f781bacd688328&sha256=767f5494bc575005b9c473dbce27feecc8eeb07d9432123830e664df93bc7576&sha1=f7072253319fea8f33a122fdf3403ca781f0a96c

Fig. 5. Example of QR code and its content.

and their time of timestamping, as well as the receipt number. The decision-maker can then compare the hashes and ensure that the evidence has not been altered since the time presented.

The structure of the secure e-CoC ledger is composed of blocks, each one corresponding to a record of an evidential item. As depicted in Fig. 3 through Fig. 6, the blocks are created from top to bottom. Each one is linked to the previous one, with the hash of the previous block.

Fig. 4 shows an example of two blocks. They are stored in the e-CoC ledger with various attributes recorded. The first attribute is the previous_hash, which plays the role of index of the block. It allows the block to be linked with the previous one, as long as it corresponds to the hash of the previous block. Then, salted hashes are written in hexadecimal form. After that comes the timestamp of the hash values, then the receipt number which is a unique identifier of 12 hexadecimal characters. The public attribute is used to indicate which blocks have been sent to a public blockchain. Indeed, some blocks will be sent to one or more public blockchain's ledger. This establishes a public anchor point, which is inalterable by the trusted entity that manages the secure e-CoC ledger. A block can be sent to a public blockchain according to a specific time (each day for example) or according to a specific size of the private ledger (each 10 blocks for example). This attribute, if the block is sent to a public blockchain, will contain the name or an identifier of the blockchain used, and the ticket number to retrieve the value from the public blockchain. The value sent to the public blockchain is the previous_hash attribute of the block. The exact process of the interactions between the secure ledger and public blockchain ledger is described in Bossuat-Pahud and Jaquet-Chiffelle [*2018*]. In Fig. 4 for instance, the second block was sent to a public blockchain. Finally, the hash_bloc attribute is the hash value (calculated with SHA-256 algorithm) of all the other attributes of the block (in the dashed square in Fig. 3). This value will be the value of the first attribute of the next block.

As explained above, a QR code with an URL is generated for each block in the e-CoC ledger. This URL contains the non-salted hashes of the evidence, the salt, and the index of the block (the previous hash attribute). When someone scans the QR code as shown on Fig. 5, a webpage is displayed that contains details about the integrity and the timestamp of the associated hash values. For this operation, the secure ledger looks for the block which has as previous_hash attribute matching the value stored in the QR code.

From the block of interest, a search is performed in subsequent blocks to determine which one has been recorded in a public blockchain. Once the secure ledger finds such a block, it compares the value stored in this block with the corresponding value retrieved from the public blockchain ledger. This permits a public verification that the value of the block sent has not been altered. If no such block

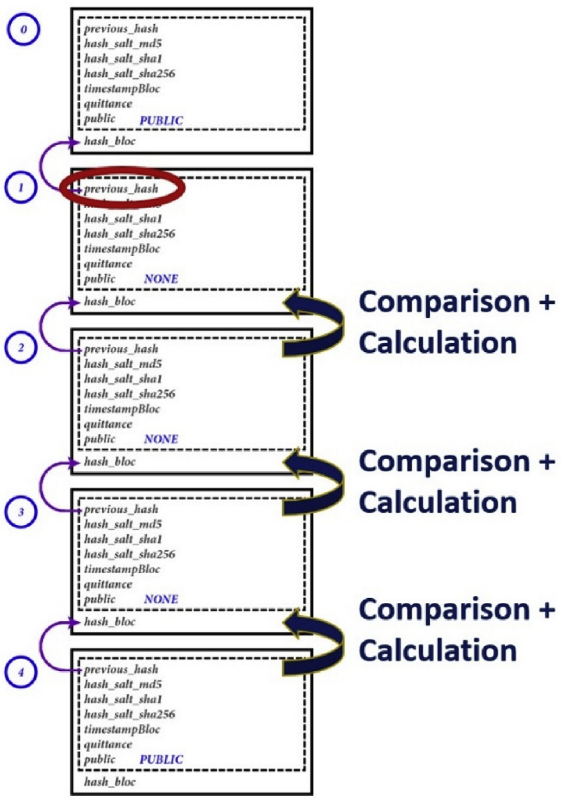


Fig. 6. Verification of the integrity of the block number one.

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```
>python aff4.py -c skype_photo.aff4 Skype-20200319-192629.jpeg -b blockchain.config

Creating AFF4Container: file://test_aff4\skype_photo.aff4 <aff4://9fccb177-a4ac-4427-
873b-3418947765fe>

        Adding: test_aff4\Skype-20200319-192629.jpeg

Sending hash values of ['test_aff4\\Skype-20200319-192629.jpeg'] to the blockchain...

Hash added to the blockchain!

The PDF receipt Registre_Quittance_1422-430E-B6C1_2020-03-28_183320_(CET_+0100).pdf was
created!

The URL to verify the integrity of the block is:

       b'127.0.0.1:8000/search?p_hash=2e38d8ba5cce9a3c69e1bd13d9684e8d6591915bf75d82958
c85218ca51a6b88&salt=6c21eaeec8da48a5b4c7ced1cdb48b33&md5=15ccf889015e686a8c42d8c6fd898
8b4&sha1=48690a0c7aba71cf2957eca6a4eedd5f0f86554f&sha256=b94b8050e161da332a38d02f40deb6
7dfdf395bdd1e36ab1cffead8e9b807255'
```

Fig. 7. Output generated by the integration of the e-CoC with AFF4 (proof of concept).

has been sent to a public blockchain, the secure ledger can still verify the integrity of the blocks, but without public verification. This is a lower level of verification, in the sense that there is no external verification, but it still allows verification. When this public comparison is complete, the verification can proceed from this point through the previous blocks: for each block, the hash value of the whole block is calculated and compared with the hash value of the block stored within the block itself and with the value of the attribute *previous_hash* of the following block.

For instance, in Fig. 6, the block to verify is the Block 1. Searching subsequent blocks finds that Block 4 has been sent to the public blockchain. The value of the previous_hash attribute of Block 4 is verified using the value recovered from the public blockchain. Then, Block 3 is verified by calculating its hash and by matching it with the value of the *hash_bloc* attribute of Block 3 and the *previous_hash* attribute of Block 4. This verification process continues for each block until Block 1.

If no problem is encountered until the block of interest, no modification was made to the e-CoC ledger and the data contained in that block can be trusted. For each hash value contained in the QR code, the system calculates the hash value of the combination of the hash present in the QR code and the salt. It can then compare the value stored in the e-CoC ledger with the new calculated value. If the values are the same, this means that the evidence has not been altered since its timestamping in the e-CoC ledger (as long as the corresponding evidence still has this hash value, of course).

If the values are not equal, this indicates that the e-CoC ledger, at least the actual block, may have been altered, or that the QR code has been altered with false hash values. Another reason would simply be that the evidence is no longer the same as it was at the time of the timestamping. The user receives a page indicating this error and the possibility to send an email to the administrator of the system.

6. AFF4 integration

To demonstrate how the e-CoC solution presented in this work can be easily integrated with existing tools, the open source AFF4 was selected [Schatz, 2015]. One advantage of AFF4 is that it encapsulates metadata which augments chain of custody [Cohen et al., 2009; Gayed et al., 2012]. In the past, AFF4 was mainly used within more mature digital forensic organizations such as Google and government entities, and required specialized tools to process [Nithesh et al., 2012]. AFF4 is becoming more widely used for both digital forensic purposes and incident response, and support has been integrated into commercial and open source digital forensic tools.

Using an Automated Program Interface (API) to the implemented e-CoC solution, any AFF4 acquisition can automatically generate an entry in the blockchain (Fig. 7) and receive a receipt in PDF form (Fig. 8). Additional metadata can be included in the blockchain, such as the unique “arn” identifier assigned by AFF4.

Specific details of the blockchain service to be used by AFF4 are provided in a configuration file, allowing users to specify their supporting infrastructure, as shown in Fig. 9.

The source code for this proof-of-work is available for reference (https://github.com/timbolle-unil/pyaff4).

7. Integration with other tools (Dc3dd example)

The e-CoC solution presented here can also be easily wrapped around existing tools, without having to modify them. For instance, a proof of concept wrapper was developed for dc3dd, a widely used forensic acquisition tool. A simple bash script was created to execute the dc3dd command and call a Python script that will send the output of the dc3dd tool to the blockchain.

Here is an example of the proof of concept in action, with its output (Fig. 10):

A receipt is also created. We also used a configuration file to set the blockchain specifications. The source code for this proof-of-work is available for reference (https://github.com/timbolle-unil/dc3dd-blockchain-wrapper). This code can be combined with other command line wrappers such as the prototype wrapper to provide enhanced chain of custody information using the CASE standard in JSON.LD format (https://github.com/casework/CASE-CLI-Wrapper).

8. Attacks and risks identified

Deliberate alteration of the e-CoC ledger

The main risk is that someone alters a stored value in the e-CoC ledger, such as the timestamp and the hashes, which would break part of the chain required for integrity verification. Specifically, when one block is altered, its integrity and the integrity of the preceding blocks cannot be verified. However, the subsequent blocks, meaning the blocks added after the altered one, are not affected by this alteration. In order to segment the e-CoC ledger and reduce the number of non-verifiable blocks when one block is altered, some blocks are sent to a public blockchain. Indeed, as the public blockchains are decentralized, it is not feasible to alter the



Fig. 8. Example of integration of the solution in AFF4 and its corresponding receipt.

```
{
    "block_chain_url": "http://127.0.0.1:8000/registreAPI",
    "case_name": "test_case",
    "filectest_aff4"
}
```

Fig. 9. Specific details provided for using the implementation in AFF4.

data. Not every block is sent in the public blockchain because of the fees needed for the transactions. This process is shown in Fig. 11 where Block 3 is altered. The integrity of the following blocks (4, 5, …) can be verified. The integrity of the previous blocks (2 and 1) cannot be verified. Because the *previous_hash* attribute of Block 1 was sent to a public blockchain, it is still possible to verify the integrity of Block 0.

The administrator still has the option to verify in earlier backups of the e-CoC ledger if the values are the same.

For example, in Fig. 12, an attribute of the Block 1 has been altered by an attacker. That means that the integrity of the Block 1 cannot be proven; the same holds for Block 0. However, the integrity of the next blocks −Blocks 2 to 4− can still be guaranteed.

An attacker could potentially alter a block of interest, and then modify the attributes *previous_hash* and *hash* of the next blocks, until the end (Fig. 12). This could stay undetected by the secure ledger, except if a block was sent to a public blockchain. The difference would thus be detected, because the value stored in the public blockchain cannot be altered.

Thus, the e-CoC ledger can guaranteed that a block has not been altered only after a posterior block is sent to a public blockchain. There is a period during which the integrity of a block cannot be guaranteed by the secure ledger (or only without public verification): between the moment of the creation of the block and the moment a next block is sent to a public blockchain. That means the more often blocks are sent into public blockchain, the smaller this time is. The intervals at which blocks are sent can be set according to a number of block and/or according to a defined interval of time. A good idea would be to send at least a block every day.

Deleting the e-CoC ledger

This attack is similar to the previous one, but in a more extreme form. If the e-CoC ledger is deleted from the server, a new secure ledger will be created when an evidence is timestamped. This would send an email to the administrator of the system, so he would know that there is a problem. To fix this, he could restore a backup, but this would mean that some blocks, the new ones since the last backup, would be lost.

Alteration of the QR code

The QR code contains an URL and the attributes that are

```
      android@ubuntu:~/case_dc3dd/dc3dd-blockchain-wrapper$ ./wrapper.sh dc3dd
if=test_file.txt of=test_file.dd

==========================

dc3dd 7.2.646 started at 2020-03-28 15:08:35 -0700

compiled options:

command line: dc3dd if=test_file.txt of=test_file.dd

sector size: 512 bytes (assumed)

          20 bytes (  20 ) copied ( 100% ),    0 s, 0.2 K/s

input results for file `test_file.txt':

   0 sectors + 20 bytes in

output results for file `test_file.dd':

   0 sectors + 20 bytes out

dc3dd completed at 2020-03-28 15:08:35 -0700

==========================

Tool:                dc3dd

User:                android

Last Login:          2020-03-28T10:54:00.Z-0700

Shell PID:           11811

Tool Location:       /usr/bin/dc3dd

Number Args:         2

Command run:         dc3dd if=test_file.txt of=test_file.dd

Start Time:          2020-03-28 15:08:35

End Time:            2020-03-28 15:08:35

Total Duration:      0:0:0:0

Sending hash values of test_file.dd to the blockchain...

Hash added to the blockchain!

The PDF receipt Registre_Quittance_E895-4C7E-9D21_2020-03-28_230835_(CET_+0100).pdf
was created!

The URL to verify the integrity of the block is:

      b'127.0.0.1:8000/search?p_hash=c71a7562280beb2edc499177b6290f6a5f4e210ff236cbf
8796464f71441f74b&salt=dd498f3732d5493f8dadb35555eff066&md5=27918d787f1afc1aeb467b53b
2fc5b61&sha1=1c32628144b252ad6fd4d1f006b6eba221193ea4&sha256=f8ba74b0c13f5745b532fde2
12fe121efda02b86aeff073baa39c68c34ecb019'
```

Fig. 10. Output generated by the e-CoC wrapper around dc3dd (proof of concept).

necessary to verify the corresponding block, i.e. the index of the block, the value of the salt and the true values of the hashes. If the QR code is altered, the e-CoC ledger would detect that the values in the QR code are not the same that were timestamped. The alteration would thus be easily detected and announced to the controller.

Automatic true validation

The true hashes are stored in the content of the QR code. Some decision-maker could think that the e-CoC ledger answers automatically true in order to support the forensic practitioner. However, the timestamp is stored nowhere in the QR code, so that's already a proof that the secure ledger has found the good block. Moreover, there is a receipt number, which is only stored in the e-CoC solution and present on the receipt, and which facilitates the human verification and assure that it was recovered from the block and not from the URL containing in the QR code.

Punctual attack

The e-CoC ledger contains no sensitive data. The hash values are normally not sensitive data, since it is infeasible to go back to the source of the hash. However, an attacker might want to know if a particular file has been discovered by the law enforcement. Thus, he would have the hashes and could search for it in the secure ledger, if he succeeds in having access to the e-CoC ledger, which should not be the case. Although such an attack would not alter the secure ledger, it should be avoided. To this purpose, the original hash values are not stored, only salted hashes are (Fig. 13). The gain of a potential attack on the secure ledger is thus reduced and the attacker is then demotivated to launch the attack.

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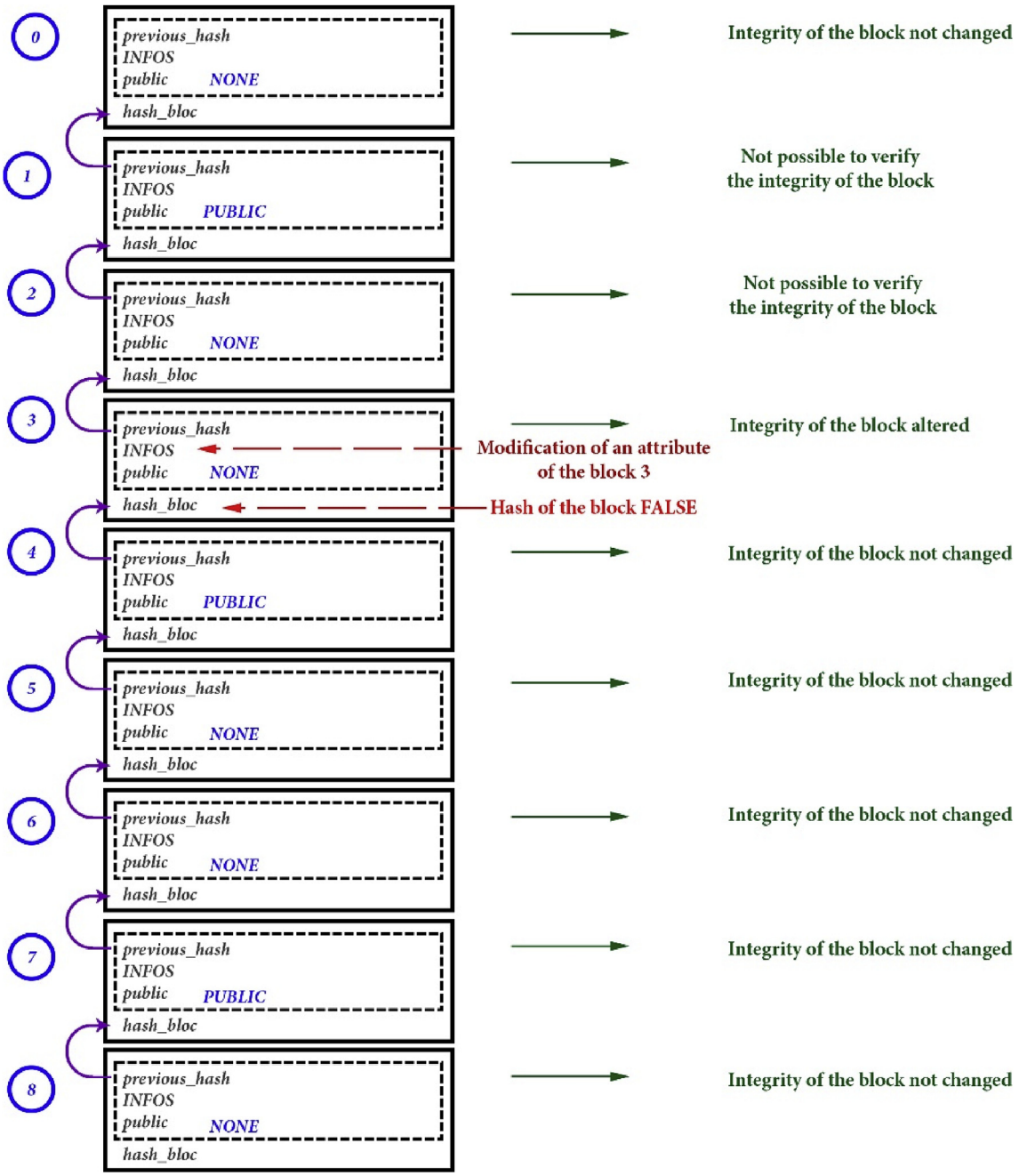


Fig. 11. Integrity of blocks after an alteration of the block 3.

The salt is only stored alongside the hash values in the QR code and de facto in the receipt. Anyone having access to the receipt has the non-salted hashes and therefore knows that the file associated to the receipt has been processed by law enforcement.

9. Future work

This section describes enhancements to the e-CoC ledger that could make it more useful.

The API could be developed to support automated verification at different steps of the analysis or in the lifecycle of the digital evidence. Users would be notified when there is a problem and when they are working on an altered version of the evidence.

Additional metadata could be added to the e-CoC record stored in the blockchain to provide provenance details such as the tools used and their versions. In this way, the e-CoC ledger could be useful when reproducing digital forensic results using a given tool or determine which evidential sources might have been effected by a bug in a particular version of a tool.

Future work reinforces the integrity verification mechanism and extends the Cyber-investigation Analysis Standard Expression for representing provenance information [Jaquet-Chiffelle et al., 2020].

Regarding the salt used and the index, it may be problematic to store it in the clear in the QR code. A possibility would also be to use private and public key, or even a system using HMAC in order to not store the salt in the QR Code. This way, the security of the e-CoC ledger would be improved. It would indeed prevent anyone to get the index of the corresponding block. Thus, an attacker who wants

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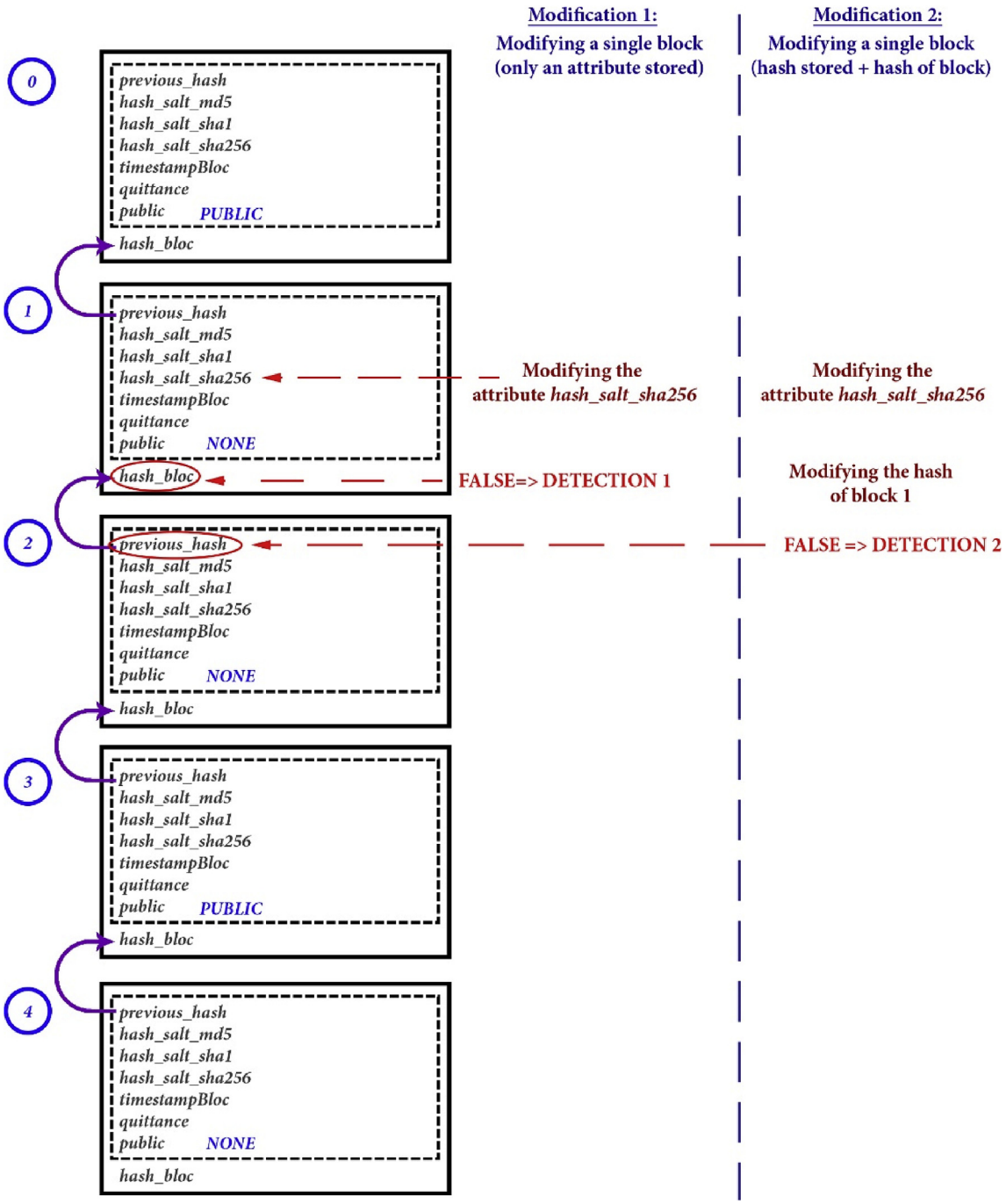


Fig. 12. Detection of the modification of data in the e-CoC ledger.

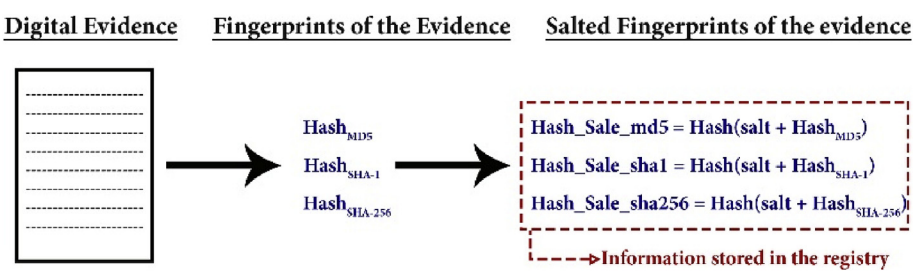


Fig. 13. Salted hashes.

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to alter the block of a specific evidence couldn't know which block of the secure ledger should be altered. The same thinking can be done for the salt: if salt is not stored in the QR code, no one can manually calculate the salt hashes and search for them in the secure ledger.

The e-CoC ledger described in this paper could be implemented using the Linux Foundation Hyperledger project, which is a Free and Open Source Software (FOSS) initiative to advance cross-industry blockchain technologies. This Hyperledger has been developed by diverse members coming from different organizations and is open source. Using this open source solution would allow anyone to implement a solution for timestamping hash values without beginning at zero and hosted by a trusted entity. The Hyperledger Composer may be more robust and secure than the secure ledger developed in the present work. One further benefit is that the Hyperledger Composer is modular, making it easy to add new hash algorithms or other information to store in the e-CoC ledger. Moreover, its nature offers interoperability, which is a significant benefit when several e-CoC ledgers are hosted by different trusted entities, enabling them to communicate with each other [Linux Foundation, 2018]. Lone and Mir, (2019) proposed that Forensic-Chain use Hyperledger Composer.

10. Conclusion

In cyber investigations, the chain of custody of digital evidence is a fundamental concern. The use of a hash value is not sufficient to guaranty the integrity of the evidence, because it says nothing about the when the hash was computed and, therefore, could be altered afterwards. The solution presented in this work provides a mechanism to timestamp hash values of evidence through an e-COC ledger managed by a trusted entity. This e-CoC ledger uses blockchain technology to prevent undetected alterations.

Moreover, some blocks are sent into a secure public blockchain, in order to minimize the impact of a potential alteration of one block of the e-CoC ledger, since the public blockchain ledger is inalterable.

The chronological independently verifiable e-CoC ledger developed in this work provides a framework for future development. Utilizing blockchain technology is an important reinforcement of e-COC as more legal matters rely on digital evidence. The solution presented in this paper is designed to be easy for developers of forensic software to implement, to help digital forensic examiners make digital evidence defensible in a court of law, and to enable lawyers and decision-makers to track the e-COC in a case.

Ultimately, this work provides a foundation for e-CoC that can be adapted to a wide range of contexts, can be integrated easily into existing tools, can be implemented using different technologies, and can be extended to include additional metadata and enhanced to provide new capabilities.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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